# PARISHRAM 2.0 2026 (Class 12th)

## CHEMISTRY

DPP 01

## d & f-Block Elements

**Q1** Which element do you expect to have the smallest atomic radius?
(A) $\mathbf{Sc}$
(B) $\mathbf{Zn}$
(C) $\mathbf{La}$
(D) $\mathbf{Hg}$

**Q2** Zr and Hf have almost equal atomic and ionic radii because of
(A) Diagonal relationship
(B) Lanthanoid contraction
(C) Actinoid contraction
(D) Belonging to the same group.

**Q3** The correct order of metallic radii is
(A) $\mathbf{Sc > Ti > V > Cr > Fe}$
(B) $\mathbf{Ti > V > Sc > Fe > Cr}$
(C) $\mathbf{V > Sc > Fe > Cr > Ti}$
(D) $\mathbf{Fe > Cr > V > Ti > Sc}$

**Q4** Due to lanthanoids contraction:
(A) $\mathbf{Fe, Co, Ni}$ have equal size
(B) $\mathbf{Zr}$ and $\mathbf{Hf}$ have equal size
(C) All f-block ions have equal size
(D) All isoelectronic ions have equal size

**Q5** **Assertion [A] : Atomic radii from Cr to Cu is almost similar.**
**Reason [R] : Sheilding effect of entering electron in 3d-orbital decrease repulsion forces of 4s-orbital electrons towards nucleus.**
(A) [A] and [R] both are true, [R] is correct explanation of [A]
(B) [A] is true, [R] is wrong
(C) [A] and [R] both are true
(D) [A] is wrong, [R] is true

**Q6** Reason of lanthanoid contraction is:
(A) Increasing nuclear charge
(B) Decreasing nuclear charge
(C) Decreasing screening effect
(D) Negligible screening effect of $f$ orbitals

**Q7** In $\mathbf{3d}$ transition series, if nuclear charge increases, the screening effect
(A) Increase
(B) Decrease
(C) First decreases and then increases
(D) First increases and then decreases

# Answer Key

| | |
|---|---|
| Q1 | B |
| Q2 | B |
| Q3 | A |
| Q4 | B |
| Q5 | B |
| Q6 | D |
| Q7 | B |

# Hints & Solutions

Note: scan the QR code to watch video solution

**Q1 Text Solution:**

- Atomic radius decreases as we move from left to right due to increased nuclear charge.
- The atomic radius increases as we move down because new shells are added.
- Therefore, $\mathrm{Zn}$ will have the smallest radius in the transition elements.

**Video Solution:**

**Q2 Text Solution:**

Due to poor shielding by the intervening 4f electrons, the atomic radius decreases significantly across the lanthanoid series. This "lanthanoid contraction" causes the almost identical sizes of zirconium (Zr) and hafnium (Hf), despite being in different rows of the d block, hence they exhibit nearly the same atomic/ionic radii.

**Video Solution:**

**Q3 Text Solution:**

The correct order of metallic radii is

$\mathrm{Sc} > \mathrm{Ti} > \mathrm{V} > \mathrm{Cr} > \mathrm{Fe}$

**Video Solution:**

**Q4 Text Solution:**

$\mathrm{Zr}$ and $\mathrm{Hf}$ have equal size

**Video Solution:**

**Q5 Text Solution:**

[A] is true, [R] is wrong

**Video Solution:**

**Q6 Text Solution:**

Negligible screening effect of $f$ orbitals

**Video Solution:**

**Q7 Text Solution:**

First with increase in the nuclear charge, the screening effect decreases since electrons are added in penultimate d-subshell which is diffused. But once the pairing starts, the increased nuclear charge is balanced by the

increased screening effect due to paired d-subshell electrons.

**Video Solution:**

Android App | iOS App | PW Website